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Roll No

CE-8001 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Advance Structural Design -II (Steel)
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume data suitably.

1. A simply supported riveted plate girder spans 13m and carries a uniformly distributed load of 80 kN/m inclusive of self-weight of girder. The plate girder consists of 1600 mm × 12 mm thick web, plate, 2ISA 200 mm × 100 mm × 15mm flange angles and 500×12mm cover plates in each flange. Design end bearing stiffeners. 14
2. Design a railway bridge for following data: 14
Type of bridge - Deck Type Plate Girder Bridge
Span - 16m between centres of bearing
Gauge - Broad, Single track, main line
Distance between centres of plate girders = 2 m
3. Design a rectangular steel overhead tank to hold 1,50,000 litres of water. The height is 2.2m and depth of water is 2m. Steel plates are available in 2m and 1.25m width and of any thickness. Design: 14
 - a) The bottom plate
 - b) The bottom tee cover
4. Design for Delhi, a self supporting steel stack of height 110 m above the foundation. The diameter of the cylindrical part of the chimney is 4.75 m. The foundation has to rest on soil having capacity 180 kN/m³. The topography at the site is flat and location is of terrain category 2. Design any three sections along with foundation. 14
5. What do you mean by Transmission line tower? What are design procedure and specifications for the design of Transmission line tower? 14
6. What are the types of steel stacks? Explain the forces acting on it? 14
7. Explain the following: 14
 - a) Shape Factor
 - b) Conditions for Plastic Analysis
8. Write short notes on: 14
 - a) Advantages and Disadvantages of steel structures
 - b) Load Factor
 - c) Flat width ratio
 - d) Stiffened compression element